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Multidisciplinary project Report

Book Store Application with Unique Book Exchange and Donation Features **Book-Hub**

Year 2022-2023

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Realized by:

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Thanks

First and foremost, we would like to thank Almighty Allah for giving us the strength to successfully complete this project.
thank every person who could ever lend us a hand to accomplish this modest work during those few months.

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Index Table

1. Introduction

1.1 Problematic 1

1.2 Objectives 1

2. Technologies used..... 3

3.Functionalities

3.1.Registration Screen5

3.3.Home Screen..... 6

3.4.Notifications..... 7

3.5.Create a new announcement..... 8

3.6.Book Details Screen 10

3.7.Check Seller Profile11

3.8.chat12

4. Diagrams 14

4.1 User Flow Diagram 14

4.2 Use Case Diagram 11

4.3.Database Schema22

5. Conclusion..... 14



1.Introduction:

We are delighted to present to you a groundbreaking project that focuses on the development of a revolutionary book store application. Our application aims to address the challenges faced by readers in accessing books, providing a platform for buying, selling, exchanging, and donating books. We believe that by incorporating unique features such as book exchange and donation, we can create a vibrant and inclusive reading community.

1.2.Problematic:

Access to a diverse range of books, financial constraints, and limited resources for book acquisition are some of the challenges faced by avid readers. Additionally, many readers have a surplus of books they have already read and no longer need, while others may struggle to afford books or have limited access to libraries. This creates a disparity in book availability and restricts the opportunity for individuals to explore new reading material. Our project aims to tackle these challenges by providing a solution that promotes accessibility, affordability, and a sense of community among readers.

2.2.Objectives:

To respond to this problematic ,our project aims to address the challenges faced by avid readers. Our objectives are designed to promote accessibility, affordability, and foster a sense of community among readers. These objectives are outlined below, considering the different actors involved, namely the readers, sellers, and individuals in need of books:



- ◆ **Enhance Accessibility:** Our primary objective is to make books easily accessible to readers, irrespective of their financial situation or geographic location. By incorporating a user-friendly book store application, we aim to bridge the gap between readers and a vast collection of books.
- ◆ **Encourage Book Exchange:** We seek to encourage readers to exchange their books with others who are looking for a change. By facilitating book swaps, we promote a sustainable reading culture and foster a sense of community among book lovers.
- ◆ **Facilitate Book Donations:** Our objective is to create a platform that enables individuals to donate books they have already read and no longer need. By facilitating book donations, we aim to promote literacy and ensure that books find new homes where they are appreciated and utilized.
- ◆ **Personalized Book Recommendations:** We strive to enhance the reading experience by providing personalized book recommendations based on users' preferences and reading history. Our objective is to help users discover new books that align with their interests and expand their literary horizons.
- ◆ **Foster a Reading Community:** We aim to create a dynamic and engaging reading community through various social interaction features. By incorporating book clubs, discussion forums, and the ability to follow favorite authors and fellow readers, we encourage interactions, sharing of insights, and the building of connections among users.

This report is organized into two chapters. The first chapter will contain a description of the technologies used to develop our mobile application . The second chapter will expose the proposed functionalities. We will close this report with a conclusion in which we will propose perspectives with the aim of making our work better.



2. Technologies used

In this chapter, we will show all the technologies used in the development of our Book-Hub application . We used several technologies according to the objectives set above.

2.1. Front-end: For the validation of tickets, we have opted for **Android Studio** development environment in addition to **Flutter** .



Flutter



Android Studio

2.2. Back-End : In order to group and orchestrate the components as well as have a well-structured database, Visual Studio, MongoDB, RESTful APIs, and Node.js were chosen as the appropriate technologies. These technologies provide a robust and efficient back-end framework for our book store application.



Vs code



mongodb



Node js

2.3.design:



figma



illustrator



BOOK-HUB

2.4.Communication tools:

Finally, communication and work synchronization between team members was assured through tools such as **GitHub**, **Telegram** and **Google Meet**.



Telegram



Google Meet

3. Functionalities(A view on the application)

This chapter describes some the features :



BOOK-HUB

3.1.Registratration Screen:

→ The authentication process is simple, just fill in the requested fields.

BOOK-HUB

Please fill your details to login.

Email or Phone number 

Password 

LOGIN 

[forgot password?](#)

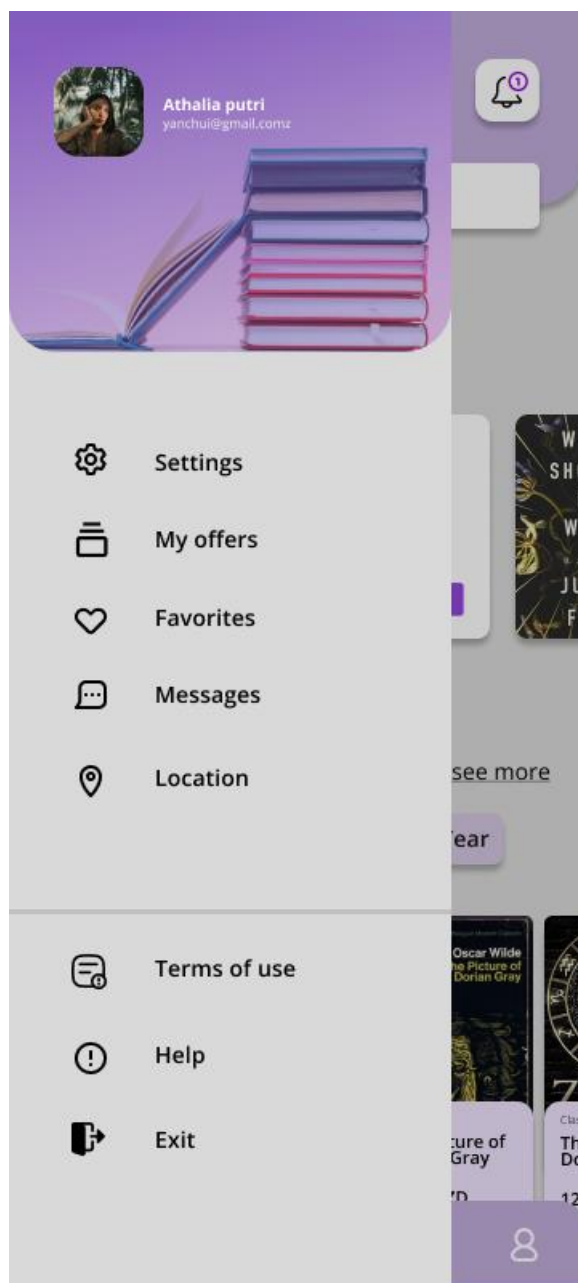
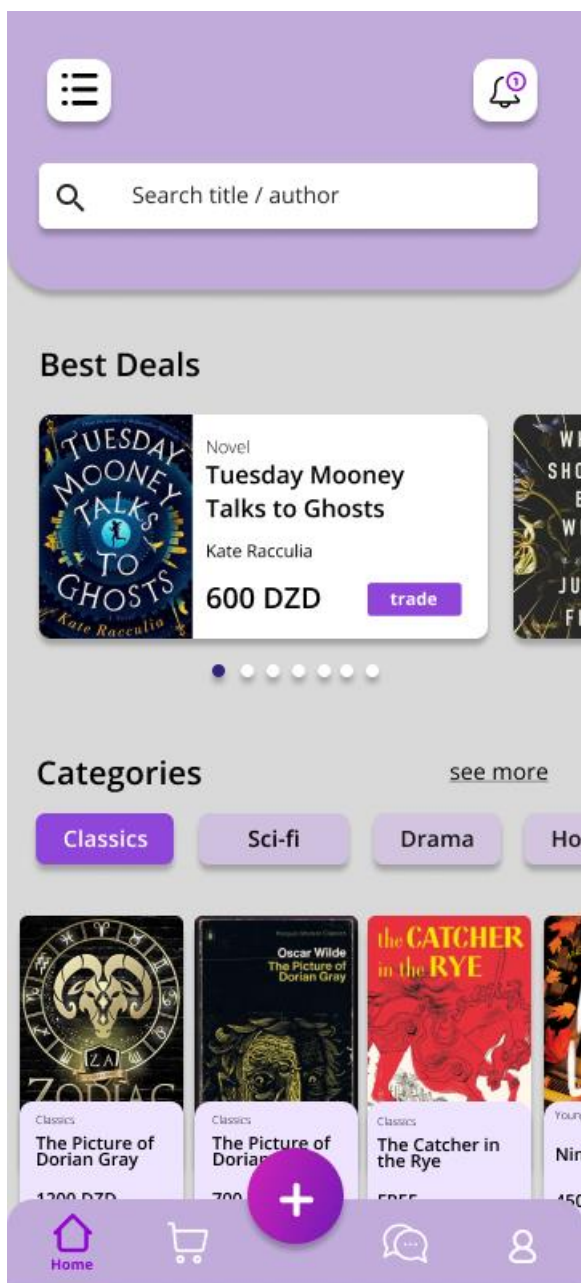
New member? [Register](#)



BOOK-HUB

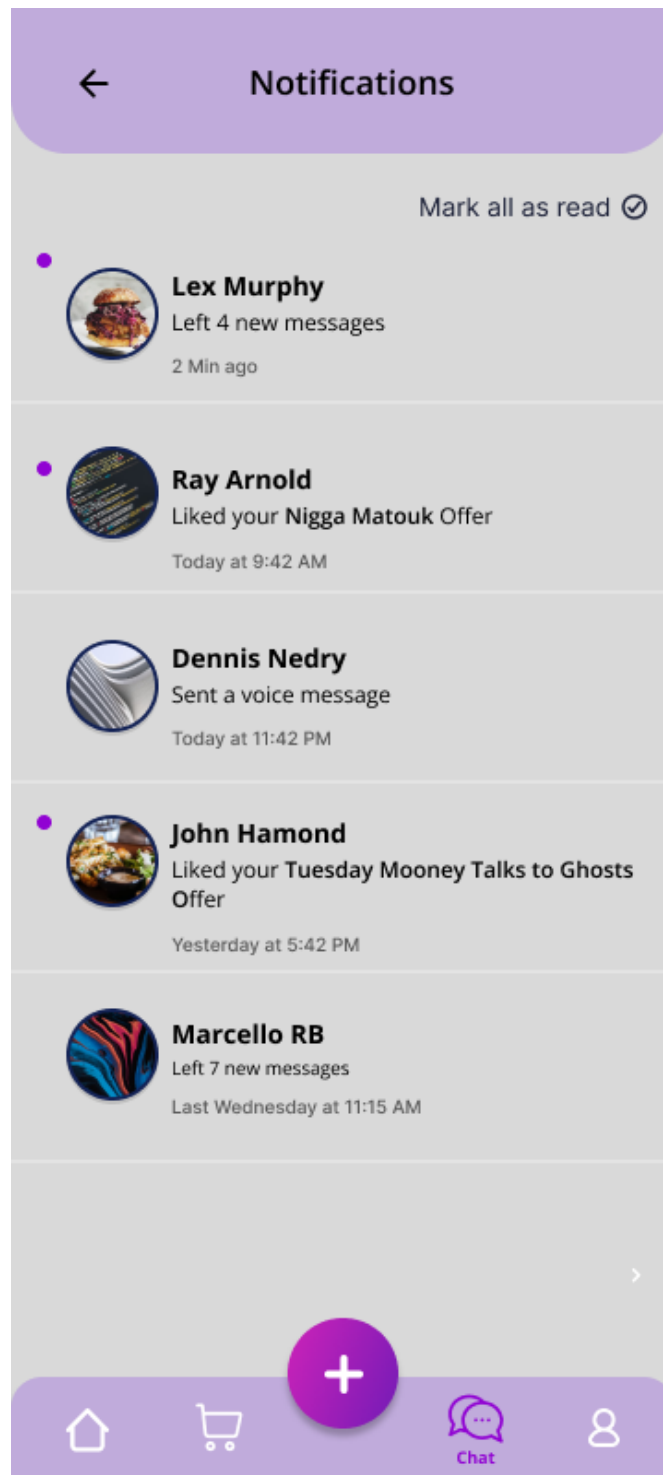
3.2.home Screen:

- a user-friendly interface that serves as the primary entry point for the book-hub application. It includes essential elements such as a search bar for easy book discovery, a menu icon for navigation, notifications for important updates, and a visually appealing section showcasing books available for sale, exchange or donate.



3.3.Notification:

The notification feature provides users with important updates, alerts, or messages. It can notify users about new book releases, special offers, order status updates, or any other relevant information they need to be aware of.



3.4.Create a new announcement :



BOOK-HUB

This feature includes the following functionality:

1.Book Details: Users can enter relevant information about the book they want to sell, such as the title, author, categories, type of selling,quantity, price,state, and a brief description. These details help potential buyers understand the book's attributes and make informed purchasing decisions.

Upload Images: Users upload images of the book's cover or relevant pictures to accompany the offer. Visual representations help attract potential buyers and provide them with a better understanding of the book's appearance.

 Create new announcement

Book details:



Insert the Books Image

Book's Name

Selling type

Category

State

Author





2. The Contact Information: Users may be required to provide their contact information, such as an email address or phone number, to facilitate communication between the seller and interested buyers. Alternatively, the application may provide messaging or chat functionality for direct communication within the platform.

Submitting the Offer: Once all the necessary details are entered, users can submit the offer.

 **Create new announcement**

Contact Info:

Full Name

Email Address

Phone Number

City

Region

SUBMIT

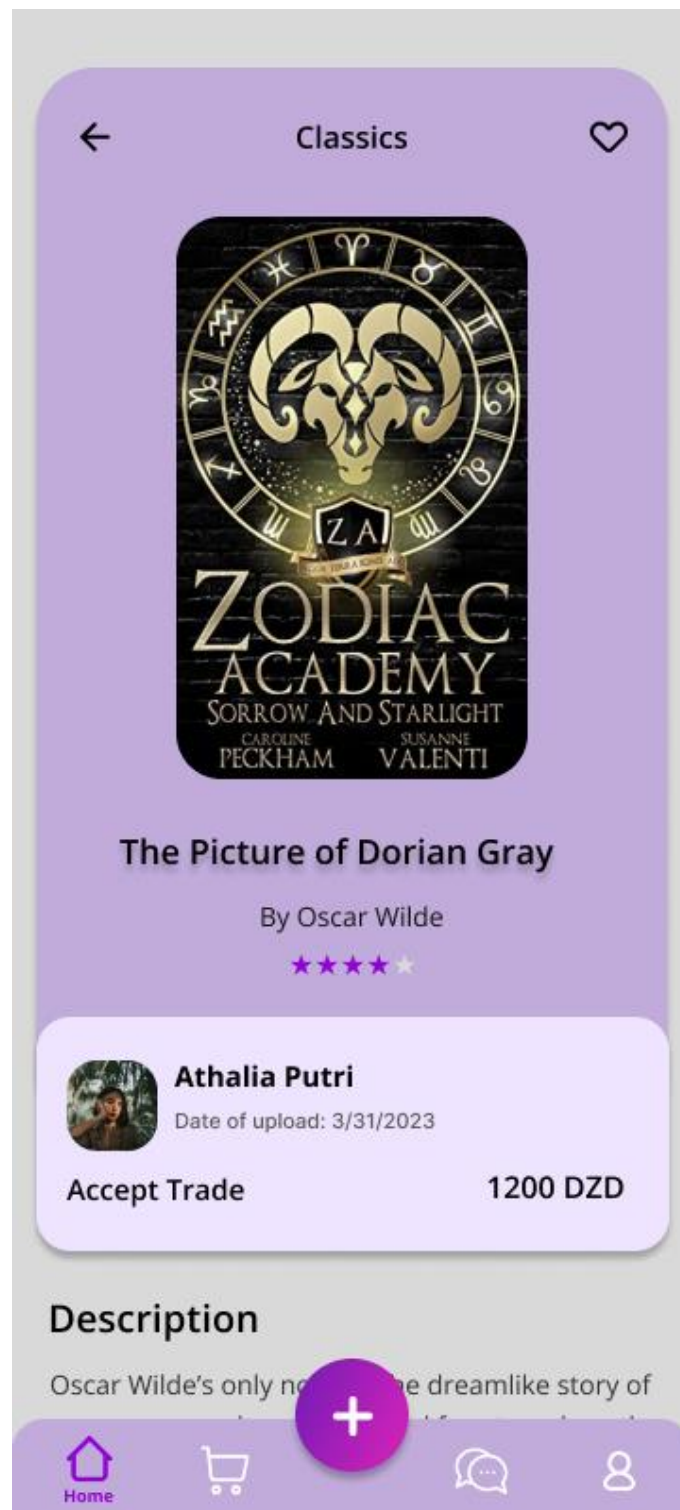
    



BOOK-HUB

3.5. Book Details Screen :

This feature allows users to access comprehensive information about a specific book. This feature enhances the user experience by providing in-depth insights into the book's content, category, pricing, and seller information...

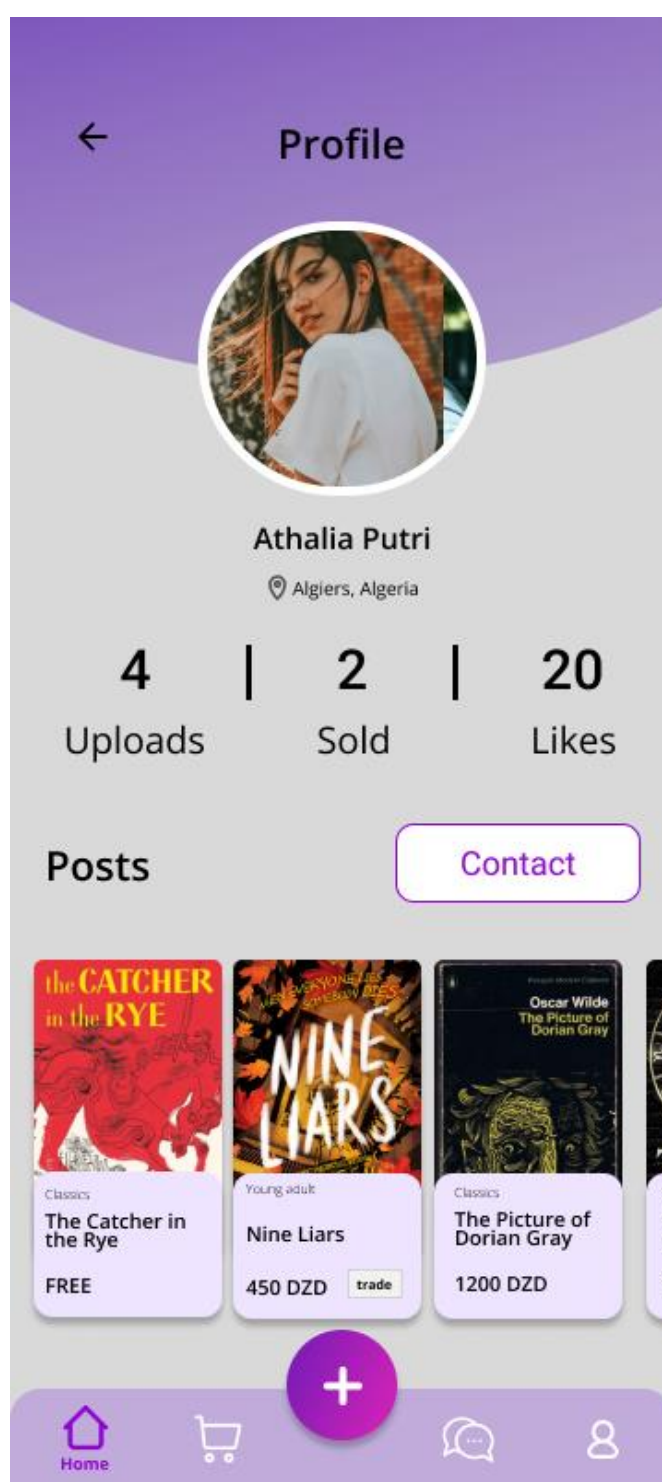




BOOK-HUB

3.5. Check Seller Profile:

This feature allows users to access information about the seller's profile within the book-selling application. It provides users with essential details, such as the seller's profile picture, name, and potentially additional information.

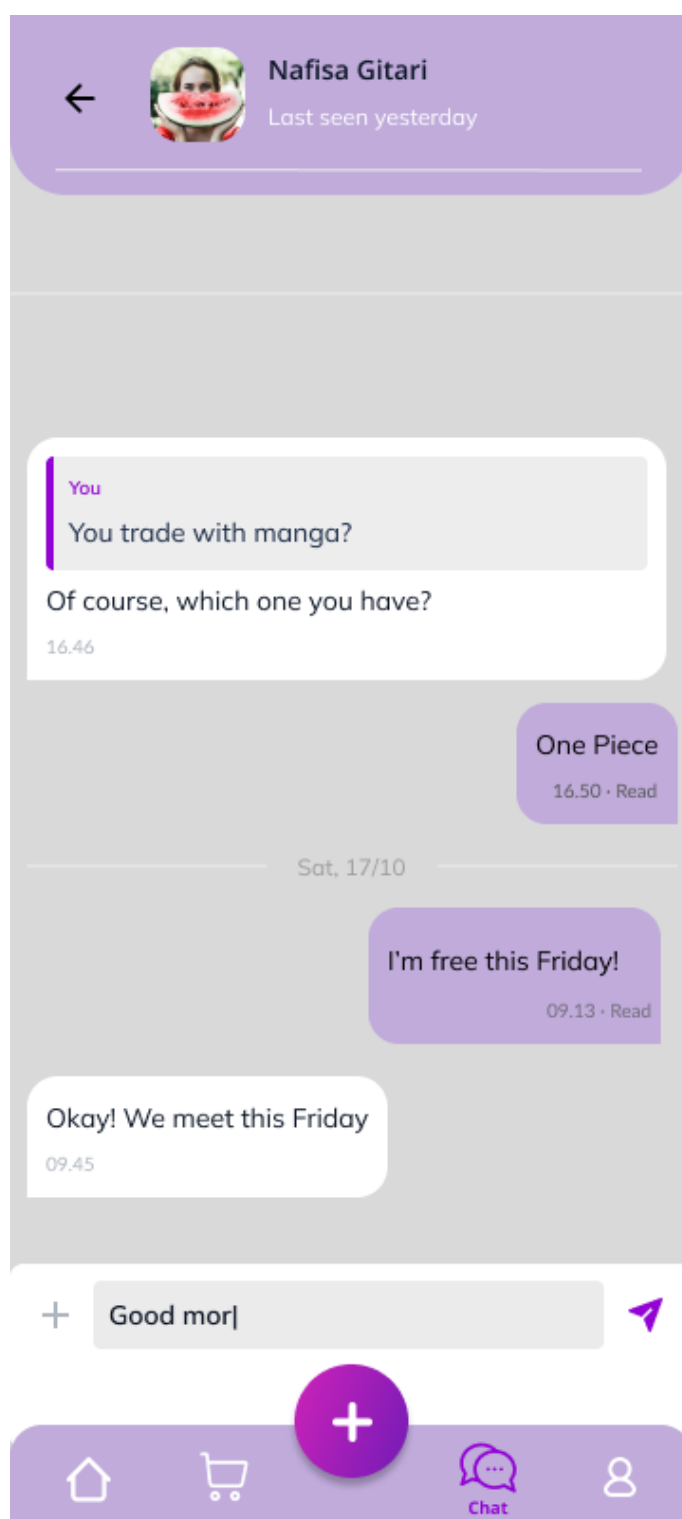




BOOK-HUB

3.6.Chat :

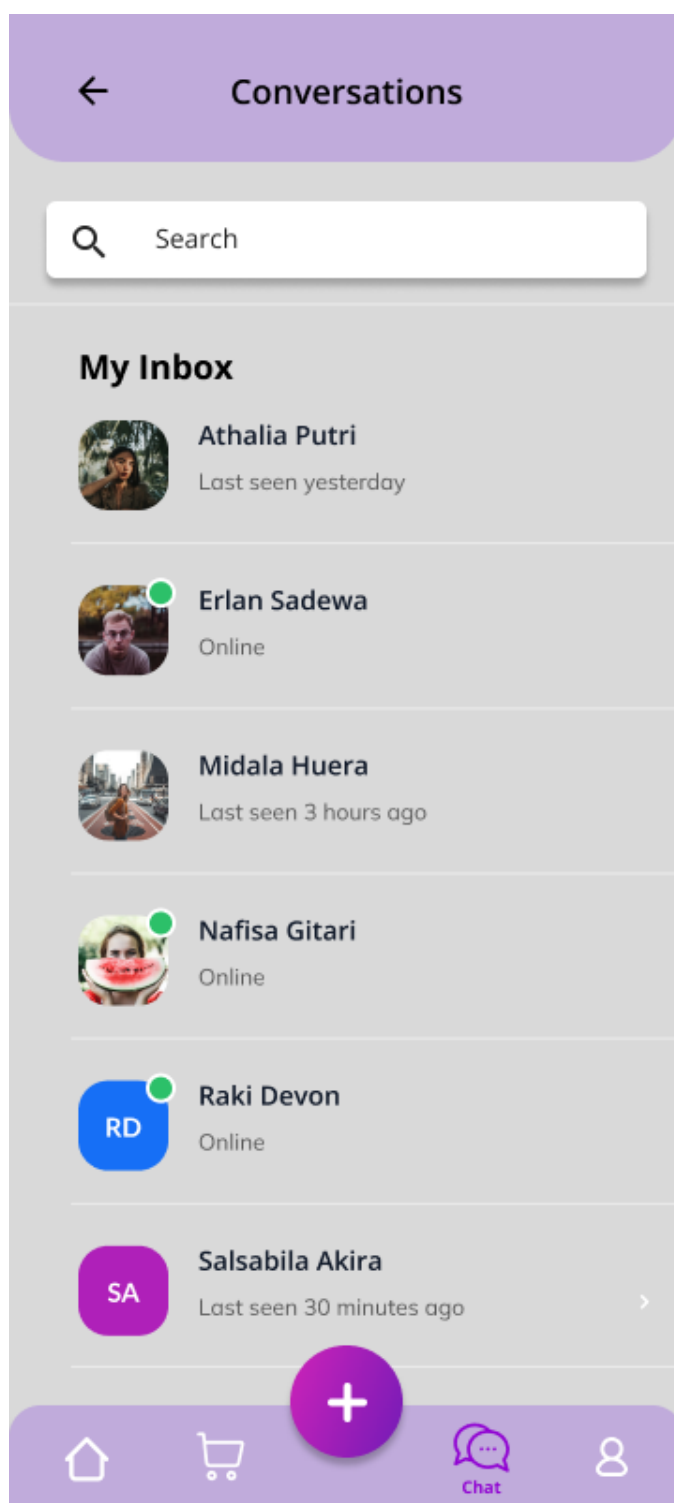
-The chatroom feature provides a dedicated space for communication between sellers and buyers. It offers an interactive and immediate way for users to engage in conversations without delays. Users can exchange messages, ask questions, negotiate prices, and discuss book-related details in a conversational manner.





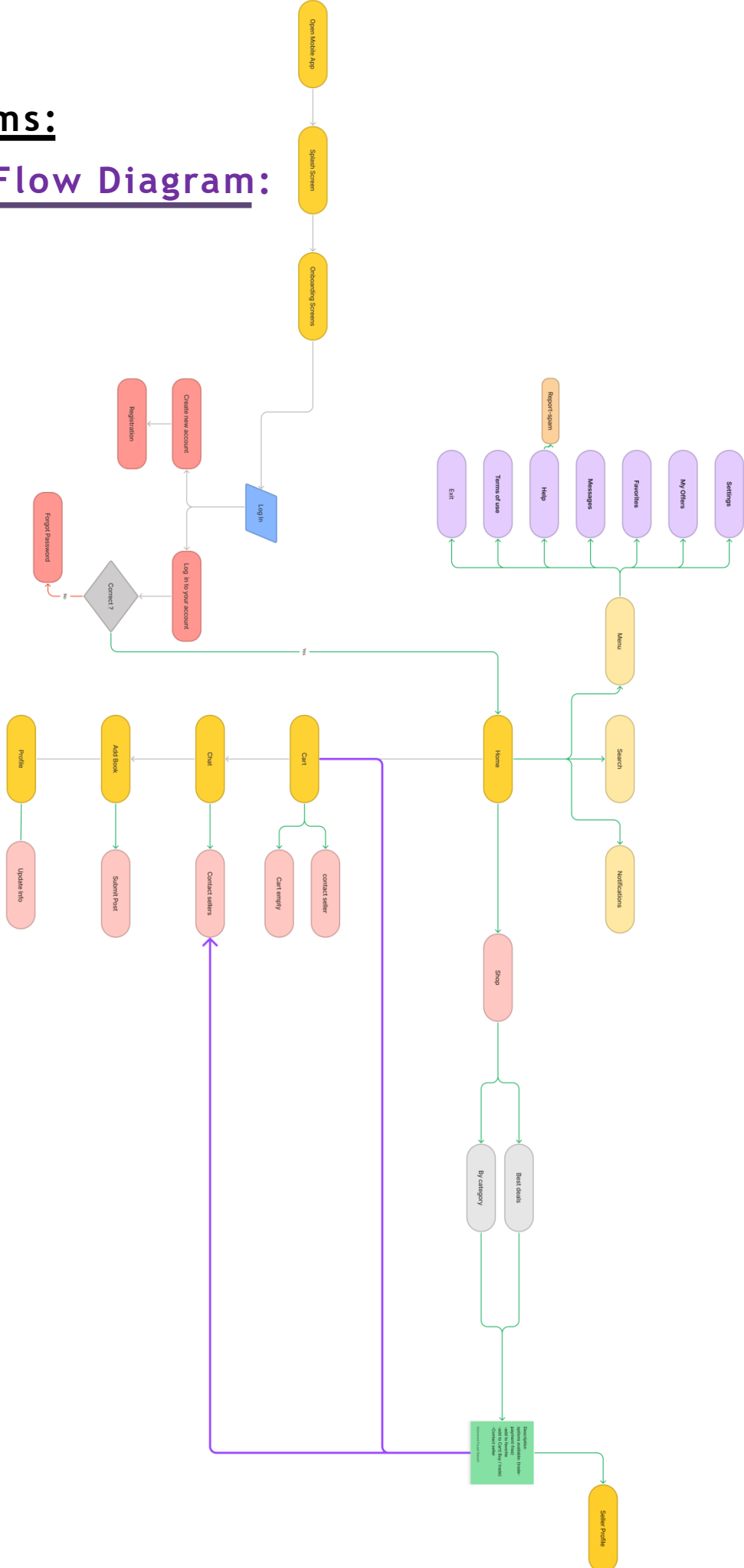
BOOK-HUB

-Within the Book-Hub application, users can access their conversations or message threads with other users. This feature allows users to view past and ongoing conversations related to book inquiries, negotiations, or purchase discussions. Users can see the participants' names, dates and times of messages, and the message history.



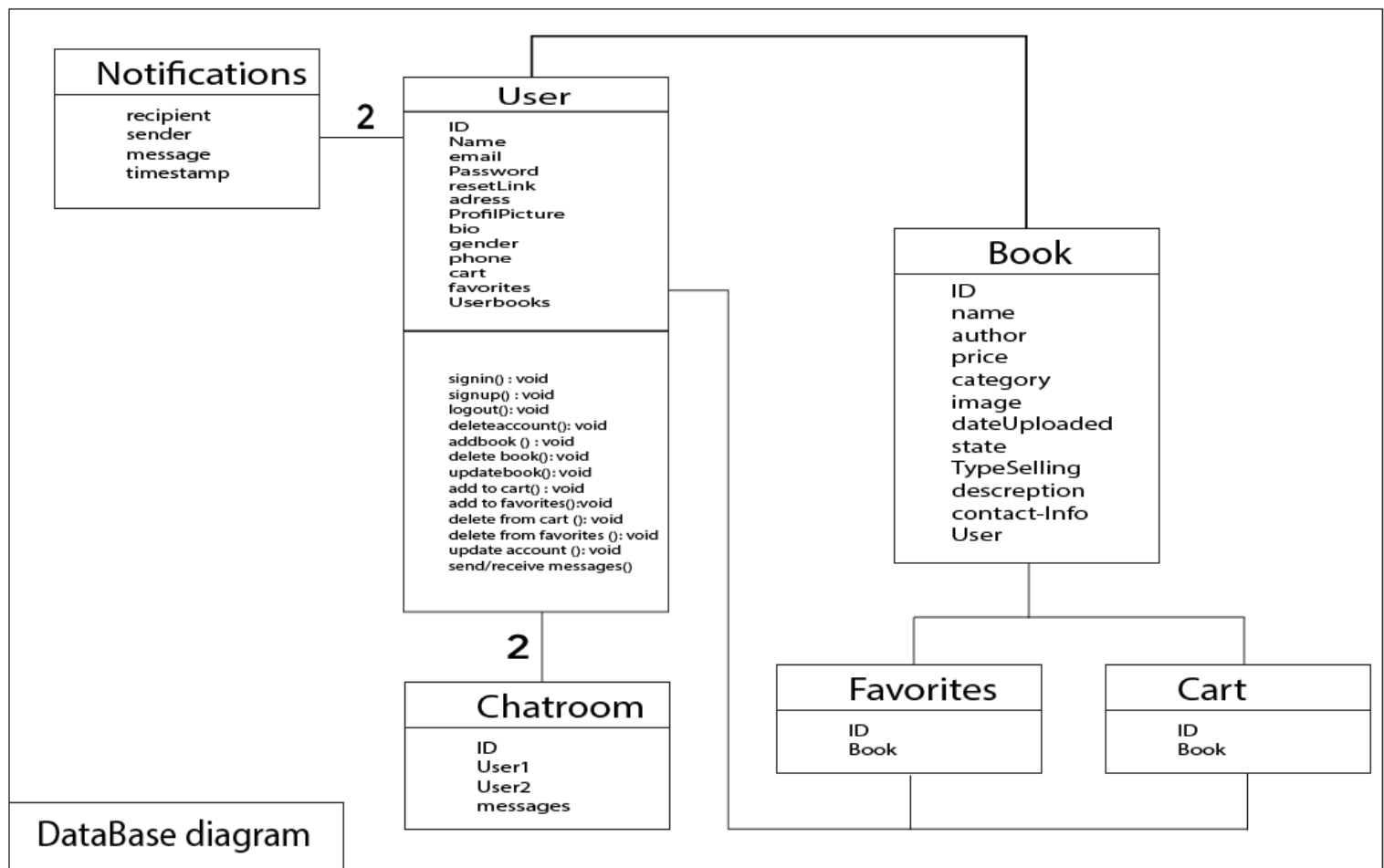
4. Diagrams:

4.1. User Flow Diagram:

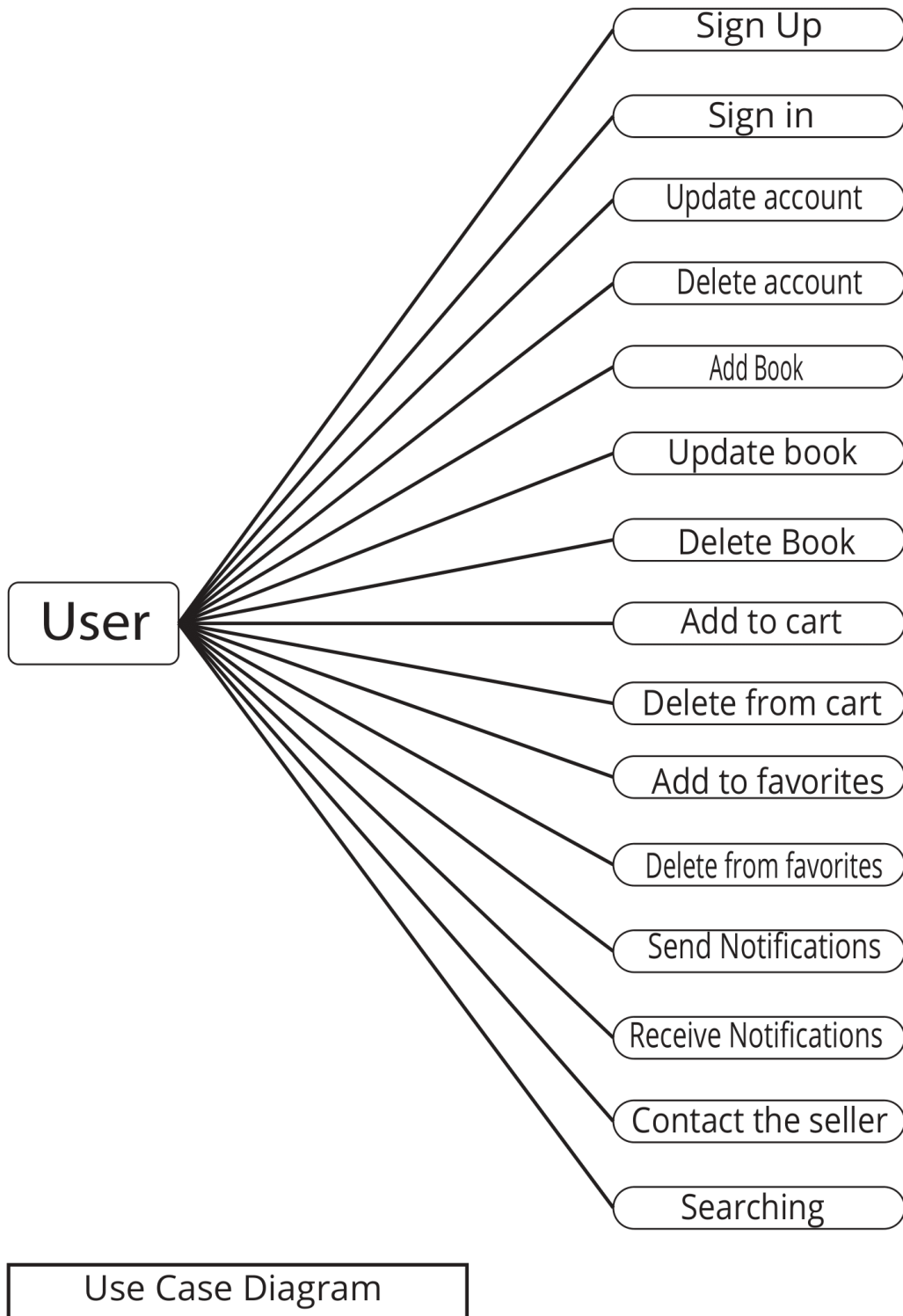




4.2. Use case diagram:



4.3.Database Schema:



5. Conclusion:

In conclusion, our book store application seeks to address the challenges faced by readers in accessing books by providing a platform that encompasses buying, selling, exchanging, and donating books. Through our unique features, we aim to enhance accessibility, affordability, and community engagement within the reading ecosystem. Together, let's embark on a journey to create a vibrant and inclusive reading community through our innovative book store application “Book-Hub”.